

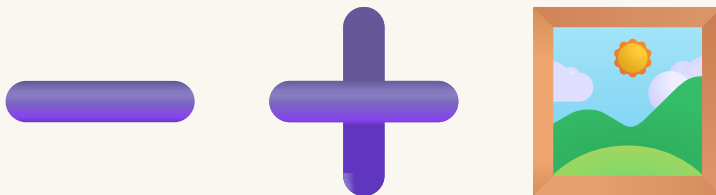
Lines & Rectangles

A loop draws a whole line at once

반복문 하나로 벽에 선을 쭉 그려요

BIG PICTURE

Four lines → a picture frame



Grow x to go across. Grow y to go up.

선 네 개를 모으면 그림 액자가 돼요

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

A loop repeats a step

Instead of placing 5 blocks by hand, a loop does it **5 times** for you.

```
x = 0
repeat 5 times:
    place stone block at (x, 0)
    add 1 to x
```

한 칸씩 옮기며 같은 동작을 반복해요

CONCEPT

Which number grows?



grow x

line goes
ACROSS



grow y

line goes
UP

x가 자라면 가로선, y가 자라면 세로선

CODE

A line across (grow x)

```
for x in range(5):  
    blocks.place(STONE, pos(x, 0, 0))
```

`range(5)` gives $x = 0, 1, 2, 3, 4$ — a line **5 blocks long**, all at $y = 0$.

`range(5)` → x가 0부터 4까지

CODE

A line up (grow y)

```
for y in range(4):  
    blocks.place(GOLD_BLOCK, pos(3, y, 0))
```

x stays **3**, y climbs $0 \rightarrow 3$. A tower-line **4 tall** standing on the wall.

CODE

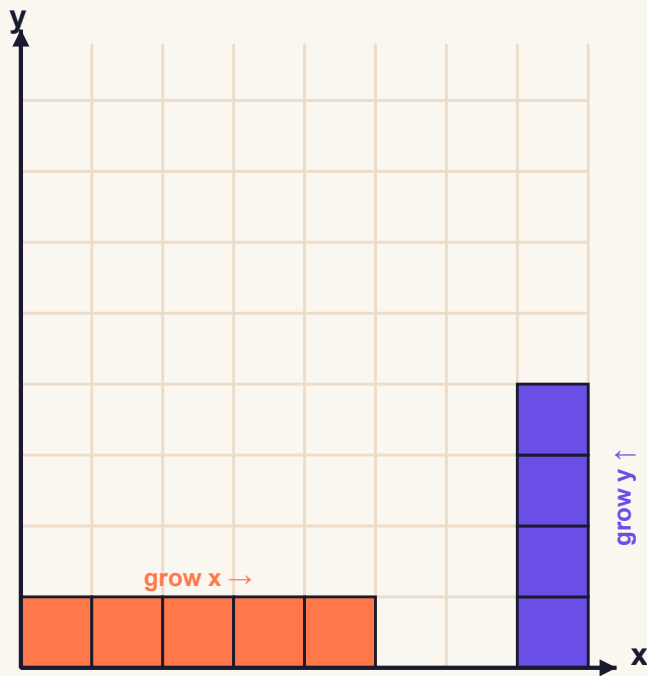
Two edges at once

```
for x in range(6):  
    blocks.place(STONE, pos(x, 0, 0)) # bottom  
    blocks.place(STONE, pos(x, 4, 0)) # top
```

One loop, two lines: a bottom edge at $y = 0$ and a top edge at $y = 4$.

SEE IT

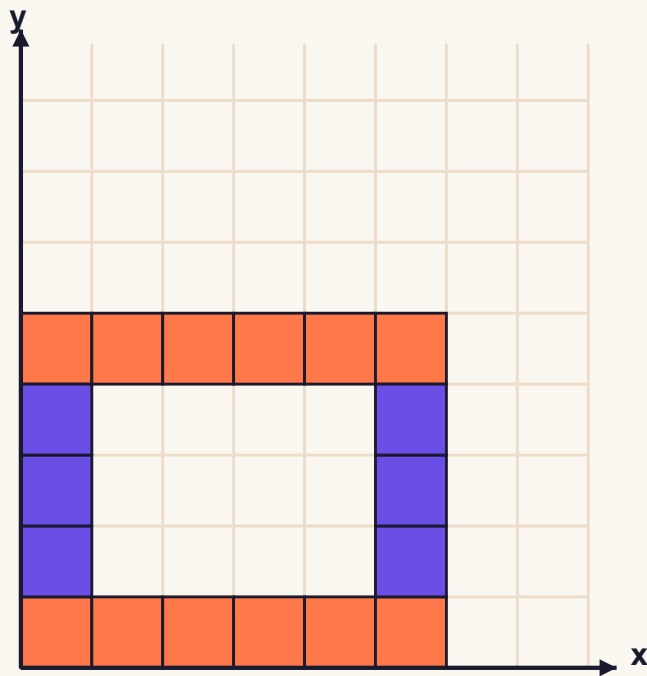
Grow x → across · grow y → up



x를 키우면 가로선, y를 키우면 세로선이 돼요

SEE IT

Four edges make a frame



위·아래 선(주황) + 왼·오른쪽 선(보라) = 빈 액자

PREDICT 🤖

How long is this line?

Predict first — then click reveal

```
x = 0  
repeat 5 times:  
  place stone block at (x, 0)  
  add 1 to x
```

Reveal output 🔍 reveal

VOCAB

New words today

loop	repeats a step many times
line	a row of blocks made by a loop
edge	one side of a rectangle frame
rectangle	four edges making a frame outline

반복문

선

변 / 가장자리

직사각형



DEBUG

What's wrong with this code?

Goal: a line of 4 blocks growing **up** from (3, 0).

buggy

```
y = 0
repeat 4 times:
    place stone block at (3, y)
```

Think first — then click next.



DEBUG

The bug

Bug: nothing grows `y`, so all 4 blocks land in the same spot. Add `add 1 to y` inside the loop.

fixed

```
y = 0
repeat 4 times:
  place stone block at (3, y)
  add 1 to y
```

YOUR TURN!

Build a picture frame

✓ Open your homework world + Code Builder

✓ Loop a **bottom** edge (grow x, y stays 0)

✓ Loop a **top** edge (grow x, y stays high)

✓ Loop a **left** and **right** edge (grow y)

✓ Send a photo of your empty frame



QUIZ

QUESTION 1 OF 3

If a loop grows **x** each turn, which way does the line go?

Diagonally.

Up the wall.

It stays in one spot.

Across the wall.



QUIZ

QUESTION 2 OF 3

How many blocks does `for x in range(6)` place (one per turn)?

7

0

6

5



QUIZ

QUESTION 3 OF 3

A loop places blocks but they all stack in one spot. What's missing?

More blocks of a different colour.

A bigger repeat count.

A line that grows the variable, like `add 1 to y`.

Nothing — that's correct.

QUIZ

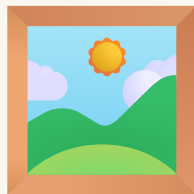
Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Frame built!

Next: fill the frame with pixel art.

다음 주: 픽셀 아트 그리기